Statistical Foundations of Latent Structure and Representation Learning

Columbia University • November 4–5, 2026

Each talk is 35 minutes including Q&A.

School of Social Work (SSW) Building
1255 Amsterdam Avenue, New York, NY 10027

Wednesday, November 4

Morning talks: **SSW C06** | Afternoon talks: **SSW 903**

8:30 am	Breakfast
8:55–9:00 am	Opening remarks: Yuqi Gu
9:00–9:35 am	**Bin Yu** | UC Berkeley Not All Interpretations Are Created Equal: Principled Evaluation of AI Interpretability
9:35–10:10 am	**Carey Priebe** | Johns Hopkins University Performance guarantees for LLMs
10:10–10:45 am	**Yuxin Chen** | University of Pennsylvania Transformers Meet In-Context Learning: A Universal Approximation Theory
10:45–11:10 am	Coffee break
11:10–11:45 am	**Jianqing Fan** | Princeton University Factor Informed Double Deep Learning For Average Treatment Effect Estimation
11:45 am–12:20 pm	**Florentina Bunea** | Cornell University Softmax-Structured Gaussian Mixtures for Distribution Ensembles
12:20–1:40 pm	Lunch
1:40–2:15 pm	**Lorenzo Mauri** | Duke University Scalable Bayesian Learning of Latent Structure: Integrating Data and Biological Knowledge
2:15–2:50 pm	**Hongtu Zhu** | UNC Chapel Hill FAST-Brain: A Flow-Aligned Spatio-Temporal Surrogate Brain Model
2:50–3:15 pm	Coffee break
3:15–3:50 pm	**Kun Zhang** | Carnegie Mellon University Causal representation learning and causal generative AI
3:50–4:25 pm	**Yihong Gu** | Harvard University Unveiling Invariant and Transferable Latent Factors Across Heterogeneous Environments via ATLAS
4:25–4:45 pm	Transition to poster session
4:45–6:15 pm	Early-career researcher poster session • SSW 214

Thursday, November 5

Morning talks: **SSW C06** | Afternoon talks: **SSW 903**

8:30 am	Breakfast
9:00–9:35 am	**Annie Qu** | UC Santa Barbara Meta Fusion: A Unified Framework For Multimodal Fusion with Adaptive Mutual Learning
9:35–10:10 am	**Jiashun Jin** | Carnegie Mellon University Estimating weak spikes with Newton and Prony
10:10–10:45 am	**Tracy Ke** | Harvard University Topic-aware Ranking with Text-estimated Context
10:45–11:10 am	Coffee break
11:10–11:45 am	**George Michailidis** | UCLA Learning Latent Representations and Structured Dynamics in High-Dimensional Time Series
11:45 am–12:20 pm	**Yuqi Gu** | Columbia University Identifiable Discrete Representation Learning: From Deep Generative Models to Latent Causal Structure
12:20–1:40 pm	Lunch
1:40–2:15 pm	**Hui Zou** | Johns Hopkins University Foe or Ally: The Role of Dimensionality in Prediction with Measurement Errors
2:15–2:50 pm	**Joseph Feldman** | Washington University in St. Louis Scalable Independent Factor Analyzers using Spectral Initialization for Evaluating Large Language Models
2:50–3:25 pm	**Gongjun Xu** | University of Michigan Beyond Vintage Rotation: Bias-Free Sparse Representation Learning with Oracle Inference
3:25 pm	Closing remarks, followed by coffee